

"Accounting for Multiscale and Conscious Psychobiological Phenomena"

48:34

all right welcome back everyone my name is Devon Bay um I'll be presenting uh accounting for multiscale and conscious psychobiological phenomena in the active inference

Paradigm so for an overview I'm going to be going over the Markov blanket trick and then critiques of the Markov blanket and address how some of the previous work of Chris Fields and Friston partially addresses these critiques um before finally going over uh problems of consciousness new approaches and theories on Consciousness and how these fit within the Free Energy and active inference and then provide some ways to update the framework uh based on this information I'm also going to uh take an epistemological perspective on modeling and I'll focus on how both the conceptual mathematical formalizations of superposition and entanglement can maybe alleviate some of the problems that are identified in the critiques so

um the Markov blanket since we are on uh limited time here um I will assume that most of the people here are already familiar somewhat with the Markov blanket or active inference framework um but I'll give a brief overview of the Markov blanket focusing on some of its mathematical Frameworks as well as some of the Assumption assumptions implicit within the the framework so um for the

mathematical overview given a state s of a model it's more blanket is the set of sufficient States within the network needed to infer that state and it essentially partitions autonomous and external States and uh is helpful in identifying which variables are conditionally independent from each other and expresses The Joint probability between external and internal States u_m as mediated by blanket States so the assumptions here is that there sparse coupling between the different states in the mark of uh in yeah in in the uh in the overall system and that there is uh somewhat of a delineation between external and Inter internal States and uh that these external internal States can be properly partitioned via the markov blanket so uh the implications here the markov blanket is metaphysically distinct from Pearl blankets uh at least in the context of active inference or for energy principle I think this was uh raised by um uh by uh boomberg or and and his colleagues um in the uh in the Markov uh in the emperor's new Markov blanket paper um and so this essentially means that they are um they are uh they differentiate themselves from Pearl blankets by nature of their application and the frisen blanket itself by nature uh takes on assumptions like the existence of external internal sensory and active States um and so this is the the main

sort of differentiation also uh the
frisen blanket is not providing a
metaphysics um at least uh at least in
my interpretation and so uh so we assume
here the existence of distinct states
which may be modeled um and this means
that there are distinct States which are
properly separable and that there are
either epistemological or ontological
grounds for doing so and so the main
problem here I think is that there is
not enough justification to argue that
the the marov blanket partitioning is an
epistemological or um you know in in a
more broader sense an
ontological um boundary and that is uh
you know causally um
correct or it is it causally captures is
able to causally cap capture causal
relationships between variables um
because the existence of sensory um of
sensory active in internal States um and
external States is something which can
be disputed and which I'll discuss later
on in this talk and so uh another
implication of the markco blank is is
that it allows the modeling of recursive
structures and Dynamics by use of the re
roral ization group for example which um
apparently allows to marov blanket to
capture multiscalar phenomenon um and
finally the question of whether the
marov blanket aderess to the beian O
razor um it would simply depend on the
choice of parameters for the specific
model and does not inherently implicate
the Marco
blanket so um critiques of the marov

blanket like I mentioned earlier the ontological critique of the Markov blanket is leveraged by both Brunberg and Raja in their respective papers although as we've seen in responses to these papers frankly depends on the ontological interpretation of active inference and uh the free energy principle and if as Kerstein in his response to Bromberg's paper claims the Markov blanket should not be interpreted literally then this problem partially goes away however the epistemological status of the Markov blanket is still under question as raised in a part of Raja's paper in so far as the explanatory power of the models under some pressure so chiefly there's a broader philosophical question about the purpose of models and this is not necessarily unique to FP or active inference or the Markov blanket but nevertheless since it affects all models these are issues are raised um are raised in in in uh in in the context of the Markov blanket as well uh next we also see that there is a mathematical uh problem with the selection of Markov blanket States um because we see that the uh for example uh Dynamic causal modeling in Neuroscience does not demonstrate that uh that models provide scientists with a plausible account of the behavior of real world biological systems since successful applications of of causal modeling does not give us any reason to believe that neural systems

plausibly for example minimize free energy or that they are arotic active inference systems and so uh this means that even though the model may be applied successfully and may predict some uh Behavior successfully it doesn't necessarily mean that the model is number one an accurate representation of what's ongoing with uh you know the internal dynamics of the system and number two that it would therefore be able to capture accurately all of the internal workings of this of the system um and so uh so yes and on this mathematical side as well we also see that uran Elon published in 2023 um a paper on the constrain disorder principle and how that would uh possi dictate neuronal function and so Elon points out that despite the stochasticity exhibited by the nervous systems operation there's not necessarily a high level of control since Randomness at the lower levels never stops and thus it dictates a paradigm of reaction not action by organisms um he uses the free energy principle but does not use it to justify a form of active inference which would be minimizing certainty but rather argues that the constraint disorder principle um uh is is more preferable where a process of balancing disorder and order is more inherent to systems operation in highly Dynamic random environments so whether all environments are highly Dynamic and random and at what scales this Randomness can be

asymptotically ignored becomes uh or becomes irrelevant is still for question however the fact that these boundaries are not clear and in fact that they arguably cannot be made clear for example via the um existence of Maxwell's demon or limits on computability Etc should point us in the direction of choosing better models and better parameters for these models so what can this tell us about uh the free energy principle in the marov blanket first it points to broader problems with the free energy principle but these still carry over um implications uh the sorry these still carry implications for the ways in which the free energy principle epistemologically treats the marov blanket mainly to delineate uh a cross in between scales and uh causal uh relationships in the later portion of this talk I'll explain why entanglement and superposition are suitable replacements for the marov blanket when they actually offer a greater flexibility without constraining the mathematical and philosophical operations and notably Chris Field's paper on the FP for generic Quantum systems makes implicit use of the same some of the same affordances offered by superposition and entanglement but takes a very specific interpretation of quantum systems and I think limits the flexibility of of such modeling tools by adhering to the holographic

principle

um so uh now onto the uh

epistemological uh status of models

which I think is important to raise here

because the because the question of

whether the marov blanket can accurately

Encompass

multi-directional uh what I'll call

horizontal Pathways vertical and

horizontal Pathways even without the

presence of clear attractor States is

still under question as is the more

General Assumption of ergodicity within

the Fe framework and so um as

experimental evidence from physics shows

um systems in general not simply at

macro or micro states are afflicted by

complex Dynamics such as replica

symmetry breaking which fall into a sort

of gray area between ergodicity and

non-ergodicity and thus render the

determination of of ergodicity versus

non-ergodicity much more complicated

even with suggestions such as local

ergodicity as outlined by ramstead in uh

in the 2018

article so um I think it's important to

outline the epist ological status of

models because um it's important for

clarifying whether the marov blanket

itself is problematic within the uh the

FP framework and whether it can

potentially be reworked so from a beijan

perspective models should want to

minimize both epistemic and alator

uncertainty where epistemic uncertainty

is the reducible part of the total

uncertainty and alator uncertainty is

basically the irreducible part active inference models therefore should minimize uh both epistemic and alator uncertainty um and represent possible model configurations and parameters via prior hypothesis space where the choice of model is informed both by where we believe the qual truth is located or at least the uh the most truthful uh model is located as well as the model constraints so here we see um in green is the ideal situation where we have an ideal selection of model with some with an ideal convergence to some truth X um and then uh the purple shows an example here where uh our prior hypothesis space does not accurately Encompass the true model and hence by extension the true model is not included in the posterior finally the pink shows where the posterior does not converge to the true model um and so this is illustrating how problems can arise in these two areas first the hypothesis base does not properly Encompass the true model X which is uh the purple or blue example and then second the posterior space does not properly converge to the true model X which is our pink uh Pink example here and so constraints on X uh depend on if you're a realist or an anti-realist so for example if you believe that there is some ground truth that is universal and um unchanging and it's you know physically uh accessible then of course the true model would be a a real true model but uh if you're more of an anti-realist then it would depend on uh

you know your hypothesis space and
you know of course whether you believe
that troops can be accurately
represented or even be accessed um and
that they're Universal or whatever so
yeah is now now the question is is the
markov blanket the proper choice of
parameter for the free energy principle
and active inference so the purpose of
the markov blanket is to provide a
statistical boundary which delineates a
system from its environment and ensures
that variables are conditionally
independent um of that certain variables
are conditionally independent of other
variables in the network and it
formalizes the belief of the prior that
the system and its environment are
separate either ontologically
epistemologically physics like
literally or even causally right so um
Columbo and Palasio actually indict the
markov blanket and the active inference
model by saying that it achieves maximal
generality for minimal biological
plausibility because these boundaries
are can be drawn sort of arbitrarily and
uh even the existence of cellular um
cellular membranes for example does not
necessarily constrain causal relations
between or cross
scales so uh notably uh Noble um argues
that organisms are open systems that
operate far from equilibrium and so
again drawing from Noble's arguments uh
Elon in the 2023 paper argues that
organisms develop numerous emergent
boundaries between levels which are not

present at the single cell level so even though they may have distinct functional purposes boundaries like the skin intestinal tract or even vastus medialis depend on exchanging matter and energy through cellular membranes and so boundary is thus Dynamic interactive not passive and cannot always be defined anatomically so for example electric gradient and ion concentration gradients are actively generated across cell membranes where interactions across boundary determine an organism's functions uh not just parts of not just the the composition of the boundary or uh or the genome so of course one way this has been explored in the free energy principle and active inference is via uh the sort of recursive markov blanket but uh we see that this sort of recursive modeling is not able to properly capture um multiscale relationships and interactions across in between boundaries where essentially we would have uh you know um markov blankets that are um that are causally implicated in in more than one uh in more than one way that that can sorry in more than one way that can be described by simply the repertoire of active external um internal States uh and sensory States and so this is sort of where the case for entanglement comes into play because uh entanglement is uh is has been argued for as a way of causally representing relationships between variables in uh in

biology so first of all we see that organizational boundaries of living systems are open and flexible such that these boundaries need not be Co extensive with the organism's bodily boundaries number one so again I guess this is just to drive home the point uh that the Markov blanket should not be taken literally um but we also see that biology is inherently Quantum so uh so these um so these uh these processes of uh cellular differentiation for example uh are under are fundamentally underlied by or sorry under fundamentally you know determined by or perhaps uh sort of controlled by uh Quantum processes that dictate you know um what chemical bonds the cell is able to break for example or how certain concentrations of of ions for example flow across um across the cellular membrane and the permeability of the membrane as well can also exhibit Quantum effects so as you know the field of quantum biology grows I think we will increasingly see that the quantum levels is extremely important for dictating how uh organisms evolve and especially if you know um Quantum Consciousness theories continue to gain support uh we could see that uh Quantum processes may also be implicated in Consciousness and so causal entanglement addresses a sort of asymmetry um between what were previously considered linear processes um and um the sort of asymmetry across in between scales so uh I basically argue that causal entanglement is a

counterintuitive form of aam's Razor
because when we have entangled variables
we don't need to always explicitly
represent the dependencies of these
variables um with each other so um uh
that means that we are we can you know
in an infinite dimensional Hilbert space
or a finite dimensional Hilbert space
represent the variables uh in a way that
does not necessarily dictate uh specific
boundaries or um or constraints on these
variables which may impede the ca of
modeling of uh certain phenomena so as
the scale and complexity of the model
increases representing relationship via
entanglement becomes much more appealing
and of course uh you know Quantum
Computing also offers ways to um to
speed up um uh the the computation of of
machine learning models like act like an
active
inference so uh vti actually puts
forward a case for causal entanglement
using uh
DNA and essentially we see that uh the
in lement can serve as a form of
um uh a form of of sort of outlining the
the total the sorry the different causal
dependencies in a system without
necessarily explicating them without
putting too much emphasis on the uh the
specific properties of the causal
relationships if we know that causal
certain things are causally implement or
sorry are causally implicated then we
can take the sort of rough approach of
assuming that they might be uh you know
entangled to some degree um and so

basically entanglement can serve as a method for

representing causal Relationships by conflating correlation with causation and so the um

the in the paper vei uses chromatin structures during the interface period of mitosis to show how and basically show how it can be characterized in terms of interactions between entities at uh three different levels of biological organization so essentially the the parts or angles between the linkers the uh in the chromatin and um then uh the the chrom structure or the topological invariant structure and then finally the the in the feedback loop between how the whole which is the the overall chromatin structure depends on context properties in the feedback loop uh for example via histone dilation or histone

methylation and so essentially uh we see that they argue that DNA is a total both a total and direct cause when being manipulated where it is a total cause because DNA is implicated in its in its own manipulation and it's also a direct cause because um it's uh it's an intervention on DNA will change uh will change the the you know this the functional structural properties of the DNA um with with without changing necessarily L um additional uh non lo non-local you know properties of whatever substrate surrounds the DNA and so specificity still allows linear models to capture causal relationships

in terms of you know isolating specific variables but when we increase the scale and complexity of these uh of the model for example or what we include in the model it becomes increasingly more difficult to separate out and to sort of distinguish okay where exactly does a specificity lie and so uh this takes me to this differentiation between repertoire and connectedness when it comes to causal specificity so repertoire is basically how many possibilities exist on each side the cause side and the effect side and we can essentially in in a modeling context take this to mean the number of different values that an effect variable and a cause variable can take meanwhile the connectedness is how many or how the possibilities on both sides are connected in particular relevant ways so we can see uh again in the modeling C test that connectedness is a condition for a given function connecting the values of cause and effect variables and so repertoire does not entail connectedness and connectedness does not entail repertoire which is important to note um because essentially we see that um that uh repertoire and but by splitting causal specificity into these two independent properties we may not be able to compare causal relations from a single point of view and therefore uh you know causal entanglement will may allow for the comparison of both without necessarily giving up you know one one

for the other and uh and of course uh Ru notes that you need both repertoire and connectedness in order to come to a complete causal understanding of you know whatever processes are implicated or you're examining within the system and so the the form of entanglement between uh you know repertoire and connectedness um would would allow you to express uh both of them in the system simultaneously and potentially uh you know um extract results from uh from either of them and so uh what are the practical constraints and limitations on this uh this approach well first of all we should uh view we might view it as a case similar to Quantum versus classical Computing where as the complexity class of the problem increases Quantum Computing becomes more appealing so utilizing entanglement may not be computationally feasible right now given Hardware constraints and um modeling using the entanglement May aisc uh the extraction of direct cause of relationships but um as Noble has pointed out and this previous researchers have pointed out the relationships is a very difficult task and epistemologically it's not clear whether this is even possible once the system surpasses a uh given uh given you know point of complexity where it's simp where the dependencies between uh and across scales simply become too large to um to properly uh identify you know direct cause or

relationships and so um essentially uh this this entanglement approach may also become computationally infeasible when uh we have lots of uh you know dense variables that are not sparsely coupled but I think because uh you know the Markov blanket also you know inherently relies on the sparse coupling and sparse coupling in neuronal Dynamics has been shown uh entanglement can be taken as an as a as a sort of replacement for the marov

blanket um and so I'll just uh get into some previous work involving the free energy principle in quantum mechanics so Fields uh Chris Fields integrated Ed Quantum information Theory with the free energy principle and proposed that physical interaction is a form of information exchange uh where the holographic principle can essentially be used to formalize boundaries between systems uh additionally warier investigated the potential for tensor networks to represent um generative active inference models so tensor networks of course can offer computational benefits When r on quantum computers and the tensor Network itself was originally um designed to be used uh in um in quantum mechanics um so specifically for Fields approach uh with the quantum reference frames we see it's a different way of approaching the Markov blanket um which still formalizes boundaries using the holographic principle so it's still assuming that we can properly

distinguish the boundaries between uh
you know organisms and their
environments and that these boundaries
are both causally uh necessary right so
that there's some causal impact at the
boundary uh that cannot be ignored and
that the boundary itself uh is you know
properly
distinguishable
um so now I'll just I'll go into uh some
theories of sorry the overview of
consciousness and how it's implicated in
uh both Fe and active inference and how
it fits into the proposed uh framework
for um for for uh for utilizing
entanglement so uh the minimal causal
mechanisms of Consciousness we see that
there are regions in the brain that are
implicated in Consciousness where you
know if there uh if these regions of the
brain are damaged or not functioning
properly this leads to changes in the
overall phenomenal and conscious uh
perception of individuals so we also see
that non-equilibrium macroscopic brain
Dynamics are a robust signature of
Consciousness so at some level
Consciousness uh can you know
statistically and um uh and physically
be proven to be the sort of higher order
uh um process or or dynamic that is
going on in the
brain and so uh Consciousness and the
Markov blanket so how does Consciousness
interact with the Markov blanket and the
free energy principle does the link
between phenomenal perception judgment
awareness observation Etc disrupt the

epistemological application of the Markov blanket and do competing theories of Consciousness pose problems for active inference so uh on the first question I think that when it comes to the Markov blanket and free energy principle Consciousness presents you know as I guess with all neuroscientific theories uh is the sort of uh throw a wrench in things when we are trying to uh uh consider how you know conscious States or different levels of conscious States may not be easily or neatly put into boxes of either internal or sensory States because again you're assuming that the organism has either the organism itself has some capacity to differentiate between these states and that the states themselves can be placed into this sort of uh into these these sort of categories for representation by the Markov blanket so I think this is especially apparent when we are discussing phenomenal perception judgment awareness uh observation Etc because um these types of phenomena are very very broad ranging in terms of their uh phenomenal activity and their corresponding correlation sorry their correspondence to underlying neuronal Dynamics um and of course the research there is still ongoing but uh based on you know whatever what research there is into uh awareness for example attention or observation um there there seems to be um varying sort of uh degrees of complexity and uh also degrees of of

causality when it comes to you know
which areas of the brain are implicated
in minimal
self-awareness and to what extent these
boundaries uh initially observed can be
modified um or you know modified either
internally by by the the person The
Observer Modified by some external uh
external causal Force for example uh
like anesthetics and so the competing
theories of Consciousness sorry do
competing theories of Consciousness pose
problems for active inference and I
would say depends on the mechanism of
Consciousness right so it depends on how
these theories lay out Consciousness
because they may or may not implicate
active inference for example uh theories
which are more volitional on the on the
more volitional side that attribute um
you know Consciousness to volitional
activity which in itself is uh maybe
distinct from you know uh free energy
minimizing um or uncertainty minimizing
activities then we would see that uh
that yes these competing theories of
Consciousness do or would Pro problems
for uh for active inference
and another Dynamic that I want to
explore here is uh the unconscious right
so experimental evidence supports a sort
of minimal unconscious where we can
broadly distinguish between type one and
type two thinking so type one thinking
is unconscious processing and type two
thinking is cons conscious processing
and um researchers have sort of found
that there is a form of dynamic

unconscious which sits between the conscious and the unconscious processes that are you know directly under observation or awareness and then those perhaps autonomic processes for example where uh they're they're entirely uh unconscious and so the dynamic unconscious Theory essentially posits that the uh The Observer or the the conscious agent themselves has some capacity to uh to guide their awareness to processes that were previously um previously unconscious so uh you can essentially make previously unconscious processes conscious and again the implications for the mark of blankets uh neat delineation between sensory active and internal States um I would say this this sort of frustrates uh the the um the thinking behind Okay the thing behind trying to separate out uh the uh these um these these different states right and so theories of the brain should incorporate a transition from unconscious to conscious awareness um and of course this is extremely difficult because we don't have really a solid understanding of what would what Consciousness even is what are the minimal phenomenal sort of uh requisites of Consciousness and how sort of expansive the operations of the unconscious are and whether those unconscious operations are therefore implicated in conscious processing as well for example with uh underlying cognitive biases so uh what are the implications

for the free energy principle so both conscious and unconscious processes should be modeled however the distinction between uh sensory and internal states does not do uh this uh these conscious and unconscious distinction I think it doesn't do adjust so why is that the case well first conscious processes are sometimes unconscious processes are processes that may inherently be uh be sort of semiconscious for example when someone is uh is scratching their their leg for example right the the distinction between uh conscious and unconscious processes is one that is fundamentally dynamic and therefore um we don't know whether there even is a boundary between the awareness of scratching leg and and not scratching leg and especially if that boundary is uh is sort of um is highly Dynamic for example uh when someone is slipping in and out of Consciousness post anesthesia um and so we see that unconscious and conscious processes could influence active inference in ways that are are particularly unique to them so for example unconscious processes uh could be represented as uh as you know as priors within the model um but that doesn't always necessarily have to be the case because these unconscious processes could also be interacting as um as as internal States for example or they could be uh they could be present even as active States once they are made uh made conscious um and so the

mechanisms of Consciousness uh various
uh theories which have been proposed
could provide a mathematical basis for
expanding active inference and the free
energy Principle as well so I think this
is very important to note because uh
theories of Consciousness um sorry novel
theories of Consciousness that
incorporate you know more Quantum
information side or that correspond to
um you know the electrical sorry the the
electrical um the electrical correlates
of Consciousness for example um are
extremely important for uh for sorry May
provide mathematical Frameworks for
updating um active inference or you know
potentially May provide a useful uh
mechanisms um for modeling specific
conscious processes that could then be
incorporated into uh the framework of
inference um and so see and so yeah so
uh I think I'm missing a few slides here
but uh basically the the the overall
theories and most promising theories of
Consciousness um uh that that are that
are upcoming are you know the sort of um
the the
electrophysiological and um Electrical
uh the electrical field uh sort of
theories of Consciousness which posit
that Consciousness is actually a result
of the electrical Dynamics in the brain
which of course uh you know would need
to be a case as as for for any minimally
causal um sort of uh representation of
Consciousness but the presence of
electrical fields and the implication of
these fields in Consciousness is uh you

know a particularly new Avenue of research and we see that for example firing patterns in the brain have produced uh quasi Crystal representation and and can be represented as well um in in in the Taurus uh for example so these uh higher dimensional uh uh Dynamics within the brain correlating to Consciousness or um or these sort of unconscious processes will be extremely important in um in determining you know what processes are relevant for Consciousness and how these might be implicated in uh in for example or uh thought patterns you know traversing whatever sort of dynamic landscape that exists in the brain which again may be determined by these higher order or uh these higher magnitude or dimensional structures so um overall we see that investigating the mechanisms of Consciousness I think is fundamentally necessary for uh for FN active inference because these uh structure um or have the potential D structure uh uh thought and the ways in which uh certain uh you know energy landscapes for example in the brain are uh are traversed or are um are fundamentally interact with each other and so uh the areas for future work here um would be to unpack the philosophical dimensions of active inference which would include implicit assumptions about intelligence sapience and sentience so for example you could utilize Denny Jean's conception of intelligence which um which argues for a a a uh sorry a relationship between

the model and the modeling so and sorry
the model and the modeler um as a as a
direct form of intelligence there or you
could utilize Imon Don's theory of
individuation as a more productive
method for explaining uh the boundaries
um and cellular differentiation which
I've critiqued previously uh you could
also utilize Quantum Computing to take
advantage of quantum specific algorithms
and uh to investigate uh Quantum
biology um um with an
increasing uh with increasing scrutiny
example and and discover ways in which
Quantum biology is implicated uh in you
know processes of Consciousness active
inference and in the brain
and so essentially the the the overall
Theory here when it comes
to when it comes to uh
oh when it comes to um to to active
inference and
uh and uh multiscalar phenomena and uh
and Consciousness is that we can
essentially um if we take that the
boundaries between the the system and
these systems are not actually uh
ontological or physically causal
causally implicated um and we utilize
this sort of uh causal entanglement to
represent relationships between uh you
know different parts of the system uh
Consciousness can be taken as a form of
observation along uh specific uh you
know um causal trajectories within the
system that would essentially collapse
the superposition of these entangled
states which would produce you know

specific uh outcomes and so that's a very broad sort of method for integrating um Consciousness with you know Quantum phenomenon and entanglement but I think that there are many uh many many other sorry different there are many there are different methods of integrating quantum mechanics with uh with the Paradigm of active inference for example that don't just that are that are distinct from the holographic principle and do not rely on the explicit representation and encoding of information in uh in boundaries for example um and when it comes to for example these unconscious processes as I've outlined in the abstract these unconscious processes could be represented more so by you know the minimization of uncertainty um uh via you know the free energy Principle as always ongoing processes similar to elon's argument whereas conscious and more volitional um uh actions would be represented by the sort of observation on this Quantum system Where We Are uh where we are collapsing the superposition of specific entangled States and so I think that in this in in that way entanglement offers two sort of uh sort of benefits to the active inference Paradigm the first is that it allows for some flexibility when we are talking about uh different variables and causal connections between and across scales because we can entangle two variables that are causally distinct um sorry that are that are uh scale that

are that are distinct across scales for example but which may be implicated in some uh feedback loop for example um and this entanglement can then uh represent the causal dependencies between these variables on the other hand um if Consciousness you know is a sort of quantum process then it would allow allow uh it would allow for the integration of whatever Quantum processes that are ongoing in in whatever you know Quantum Consciousness uh um so if if you believe that Consciousness collapses the wave function for example then uh then incorporating entanglement would be a mathematically feasible um technique for incorporating this uh this form of valuation where essentially the uh the conscious agent is collapsing a specific wave function and that leads to uh you know that allows them to perform uh active inference for example by minimizing uh specific uh probability densities so um yeah that's overall uh the the scope of my talk today um and yeah I hope uh everyone uh enjoyed that and uh if you have any question questions please let me know thank you

Devin great times so I'll ask one question you'll give the closing word for the live streams for the whole Symposium so you're the only undergrad presenter what would you say excited you to get into the game and what do you say to younger Learners to come um um I am mainly interested

in uh um the sort of mathematical and philosophical

um uh I

guess um ideas behind uh active

inference and um I was I'm particularly

um uh happy to see that the active

inference ecosystem has you know

thriving uh philosophical

and uh and

mathematical um Community

um so when it comes to uh

um younger

uh

um uh participants who may be interested

in active inference I would say uh you

know keep uh pursuing um what you're

interested in and um if you have any

particular areas of research or ideas

you should always um be be looking to

share them I think and uh be looking to

expand your horizons so

yeah cool well looking forward to

hearing more from you and from others so

thanks again Devon for bringing so much

together in this last Capstone

talk thank you so much for inviting me

to uh give the talk and apologies uh I

was a little bit short on time to uh to

put together the slides so uh this was a

lot uh messier than I would have liked

unfortunately but uh yeah real life is

like that all right thanks Devon peace

thank you so much